

Lecture 4

In Class Exercises



Exercise 1

- Use the substitution method to show that the following recurrences have the solution specified:

$$T(n) = T(n - 1) + n \text{ has solution } T(n) = O(n^2)$$

Exercise 1 (solution)

$T(n) = T(n - 1) + n$ has solution $T(n) = O(n^2)$

We guess that $T(n) \leq cn^2$ for some constant $c > 0$. We have

$$T(n) = T(n - 1) + n$$

$$\leq c(n - 1)^2 + n$$

$$= cn^2 - 2cn + c + n$$

$$= cn^2 + c(1 - 2n) + n$$

This last quantity is less than or equal to cn^2 if $c(1 - 2n) + n \leq 0$ or, equivalently, $c \geq n/(2n - 1)$. This last condition holds for all $n \geq 1$ and $c \geq 1$.

Exercise 2

The solution to the recurrence $T(n) = 4T(n/2) + n$ turns out to be $T(n) = \Theta(n^2)$. Show that a substitution proof with the assumption $T(n) \leq cn^2$ fails. Then show how to subtract a lower-order term to make a substitution proof work.

Exercise 2 (solution)

The solution to the recurrence $T(n) = 4T(n/2) + n$ turns out to be $T(n) = \Theta(n^2)$. Show that a substitution proof with the assumption $T(n) \leq cn^2$ fails. Then show how to subtract a lower-order term to make a substitution proof work.

Assuming that $T(n) \leq cn^2$ will fail:

$$\begin{aligned} T(n) &= 4T(n/2) + n \\ &\leq 4c \left(\frac{n}{2}\right)^2 + n \\ &= cn^2 + n, \end{aligned}$$

Because this is greater than cn^2 .

Subtract a lower-order term and assume $T(n) \leq cn^2 - dn$:

$$\begin{aligned} T(n) &= 4T(n/2) + n \\ &\leq 4 \left(c \left(\frac{n}{2}\right)^2 - \frac{dn}{2} \right) + n \\ &= cn^2 - 2dn + n, \end{aligned}$$

This is less than or equal to $cn^2 - dn$ if $d \geq 1$.

Exercise 3

- Use the master method to give tight asymptotic bounds for the following recurrence:

$$T(n) = 2T(n/4) + 1$$

Exercise 3 (solution)

$$T(n) = 2T(n/4) + 1$$

$T(n) = \Theta(\sqrt{n})$. Here, $f(n) = O(n^{1/2-\epsilon})$ for $\epsilon = 1/2$.

□ Case 1 applies, which means

$$T(n) = \Theta(n^{1/2}) = \Theta(\sqrt{n})$$